Hospital-Integrated Wearable System for Real-Time Health Monitoring with AI & Physician Dual Consultation

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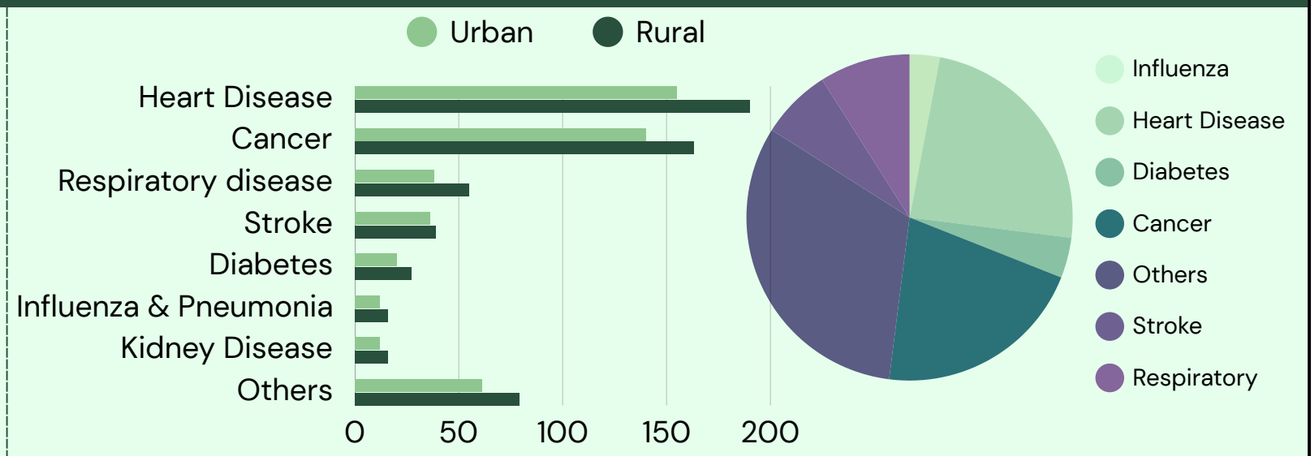
BACKGROUND

Rural healthcare systems frequently encounter challenges in maintaining continuous monitoring, making timely diagnoses, and responding promptly to medical emergencies. These issues can result in preventable complications and fatalities. Traditional care models often fail to recognize early signs of deterioration in conditions such as cardiac events, respiratory failure, or maternal emergencies.
The bar chart illustrates that rural areas have a higher death rate compared to urban populations. The pie chart indicates that rural communities experience increased mortality from non-communicable diseases, including heart disease, stroke, respiratory disease, and diabetes.



U.S. Death rates for leading causes, 2019

AIM OF THE STUDY

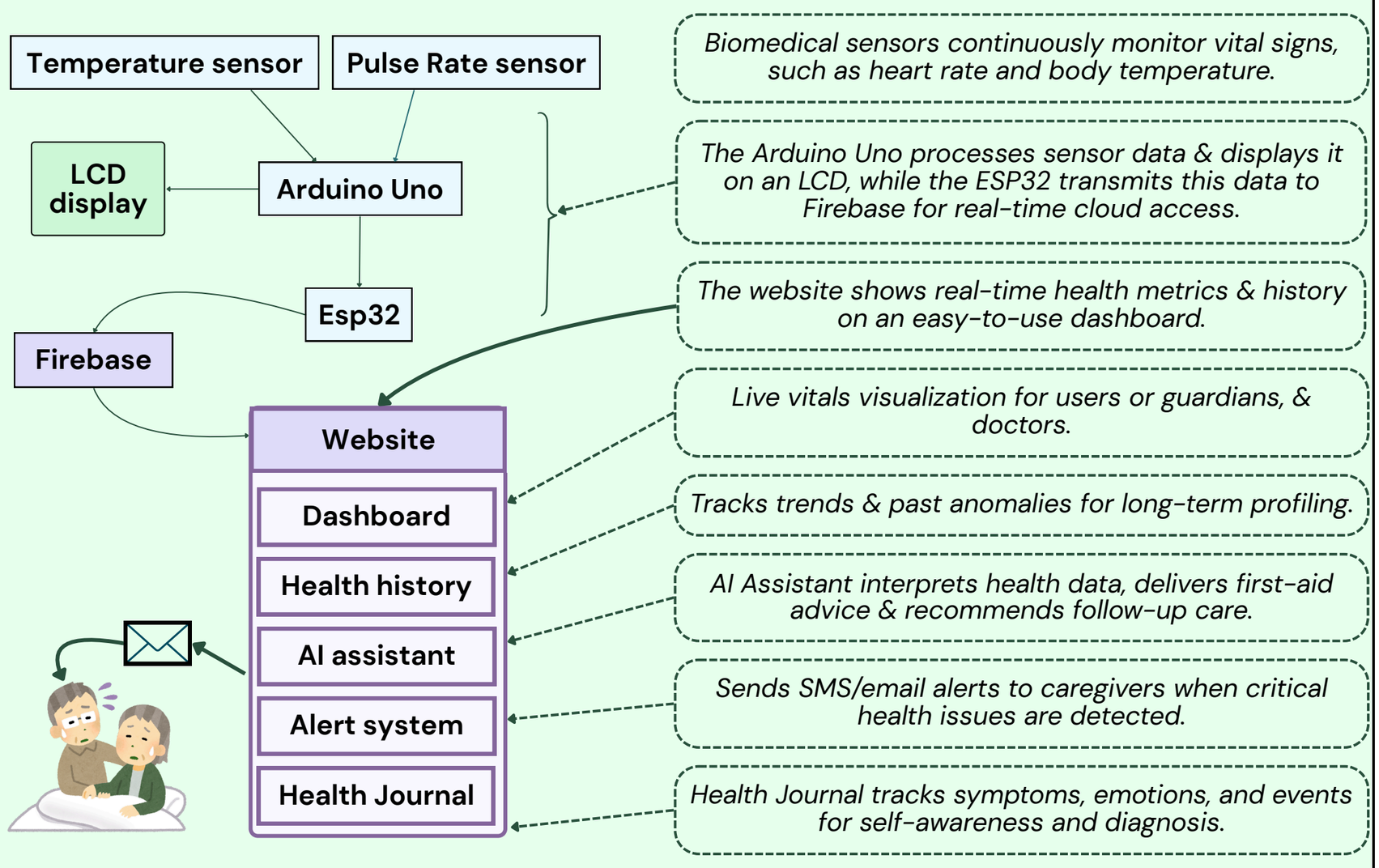
Our project aims to:

- Design and implement a low-cost, real-time AI health monitoring system using wearable sensors and cloud infrastructure.
- Enable continuous tracking of vitals with automated first-aid guidance & real-time alerts, focusing on underserved rural communities.
- Ensure accessibility and affordability to empower low-income populations with timely, life-saving health support.
- Support personalized and scalable care through health data profiling and integration of advanced biomedical sensors.

APPARATUS & COST ANALYSIS

Apparatus	Cost (BDT)
Arduino Uno	995
ESP32	450
Connecting wires	200
Switch	50
LCD display	240
Temperature Sensor	290
Pulse Sensor	420
Lithium Battery	400
Arduino IDE	free of cost
Visual Studio Code	free of cost
Firebase	free of cost
LLM API	free of cost (limited token)
Total Cost	**3045**

METHODOLOGY



RESULTS & ANALYSIS

Performance & Responsiveness

- *Tested under simulated real-world conditions*
- *Sync time: 10–15 mS (hardware → cloud → UI)*
- *Real-time vitals updated reliably via Firebase*
- *Stable data capture for pulse, temp, respiration*
- *Smooth visualization on the web dashboard*

Smart Features & Innovation

- *Tested under simulated real-world conditions*
- *Visualizes historical data*
- *Rule-based AI + NLP guidance*
- *Maps symptoms to specialists automatically*
- *No heavy ML models optimized for low-power devices*

Challenges & Future Scope

- *Lag during prolonged use → power optimization needed*
- *Stable internet required for real-time sync*
- *Privacy/security mechanisms pending*
- *Plan to add glucose & ECG monitoring*
- *Future: Better long-term performance.*

CONCLUSIONS

The system:

- Developed a real-time health monitoring prototype with biomedical sensors & cloud sync
- Enabled AI-driven decision support using the LLM API with low latency
- Features a modular architecture for upgrades and is designed for rural scalability
- Limitations include a narrow parameter range, reliance on internet connectivity, and a need for better power efficiency

This framework supports AI-integrated remote healthcare delivery.